

A CONCRETE QUESTION ABOUT FINITE OBSERVERS

Finite Directional Records as a Concrete Observer Model

Can one localized system acquire, retain, and expose a relational orientation record while its information, support, energy cost, and gravitational footprint are controlled in one semiclassical regime?

The observer-level question

Quantum-gravity observer rules give finite operational content to an observer, while de Sitter algebras include it in the construction [1]-[4]. We ask whether a localized register can make assumptions auditable by storing another system's relative orientation for later readout.

The aim is not to identify a rotor with an observer, but to expose the needed information, dynamics, localization, and gravity.

Why direction is useful

Finite spatial reference frames and their degradation are established [5]-[7]. We use a wholly relational task: if $\mathcal{O}$ is the register and $\mathcal{D}$ the target, the unknown is their alignment, not orientation relative to coordinates:

$$g_{rel} = g_{\mathcal{O}}^{(-1)} g_{\mathcal{D}} \text{ in } SO(3). \quad (1)$$

Performance is the Haar-prior global risk for estimating g_{rel} . It probes the full rotational orbit, unlike local sensitivity or Hilbert-space size alone.

The record variable

For the covariant orientation ensemble, we begin with the relative entropy of rotational asymmetry:

$$\begin{aligned} S_{dir} &:= D(\rho_{\mathcal{O}} \parallel G_{SO(3)}[\rho_{\mathcal{O}}]) \\ &= S(G_{SO(3)}[\rho_{\mathcal{O}}]) - S(\rho_{\mathcal{O}}). \end{aligned} \quad (2)$$

S_{dir} is rotational asymmetry, not classical memory, thermodynamic entropy, Hilbert-space dimension, Casimir, quantum Fisher information, or Harlow's S_{Ob} . Their possible comparison is review question 1.

Lifetime is operational: after time T , a permitted relational readout must still attain the declared risk. This is classical-record stability, not generic phase coherence.

The proposed theorem chain

Fix a class $\mathcal{C}_{dir}$ of localized register-target models, states, environments, actions, and preparation/storage/readout protocols. The intended theorem quantifies over this class:

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R_rel(T) >= F_record(S_dir, degradation, channel
error)
S_dir <= F_capacity(energy, support, gravity;
model data)
=> Phi(R_rel(T), T, support, gravity; model data)
>= 0. (3)
```

On named domains, global risk requires rotational resources; confined orbital matter turns them into support and energy cost; and a regular-bath theorem controls finite-switch degradation. Its Bunch-Davies application needs an open QFT GNS bridge.

The primary bottleneck

The QFT GNS bridge and common-action local-matter channel remain **open**. The detector calculation pays the channel-norm dimension cost but lacks a KMS GNS/Araki-Woods justification. Independently, its factorized current has a nonzero commutator for disjoint spacelike smearings, so it is not local matter. A different action must derive acquisition, storage, readout, and stress together [10].

What gravity adds, and what the sprint decided

Exact fixed-de Sitter maps define a Weyl witness [11]. Its estimator is negative, but the bounded sprint's full interval contains zero because the primal certificate is about an order of magnitude too broad. Paper R stops as a viability test: a proof limitation, not physical cancellation. The witness remains suggestive, not a theorem or capacity bound.

A supported massive Skyrmin supplies one certified matter candidate [12]. It only makes the supported-profile subproblem nonempty; the architecture must survive rejection of this realization.

Why request feedback now

The two STOPs expose the fork. We ask whether directional records are useful, whether S_{dir} can be compared with S_{Ob} , what gravitational quantity constrains the register, and whether a one-action model helps. We seek redirection, not endorsement or a publication decision.

Claim boundary: $S_{dir} \neq S_{Ob}$; record stability is not quantum coherence; measurement back-action is not gravitational backreaction; fixed-background response is not self-consistent gravity.

Finite directional records as an observer model

Dependency map: rigorous ingredients, conditional bridges, and the theorem still to be built

Task: infer the relative $SO(3)$ alignment of two localized physical systems - never an absolute direction.

STATUS

[PROVED]

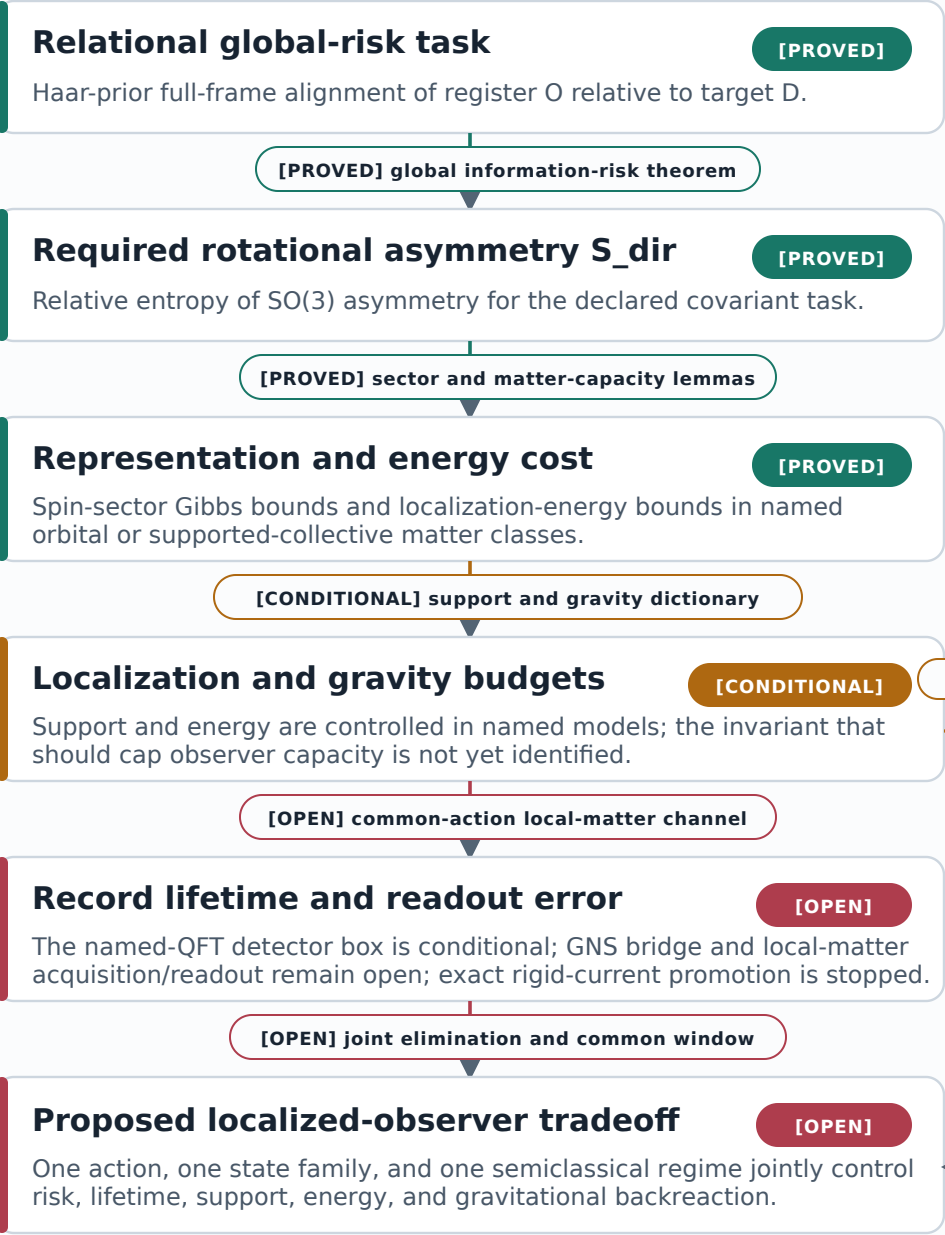
theorem on a declared domain

[CONDITIONAL]

model-dependent or floating evidence

[OPEN]

OBSERVER-REGISTER CHAIN



THE PURPOSE OF THIS PACKET

Where the observer model should connect to quantum gravity

The response sprint did not certify zero exclusion. These questions decide whether a different bridge is worth building.

- 1 Is S_{dir} , used here as an operational $S_0(3)$ -asymmetry resource for a covariant alignment task, meaningfully comparable to the observer entropy S_{ob} in the observer rule, or are these fundamentally different resources?
- 2 Which gauge-invariant quantity should replace or refine the fixed-background Weyl witness as a capacity constraint in a de Sitter patch: collapse, horizon or quantum-extremal-surface displacement, or another invariant?
- 3 Would a one-action semiclassical model deriving relational record formation, lifetime, localization, readout, and gravitational footprint clarify the observer rule, or is a closed-universe gravitational path integral essential from the outset?

Feedback target. We are not asking whether the present realization is complete. We are asking whether this is the right operational bridge to build, which quantity should replace the provisional gravitational witness if needed, and which observer-level identification should remain deliberately absent.

Auditable Status of the Directional-Record Program

[PROVED] means an analytic or computer-assisted theorem on the stated domain. **[CONDITIONAL]** means an exact implication under a stipulated effective model or source-bound floating evidence. **[OPEN]** means a missing implication, norm estimate, or physical identification. A composite claim inherits the least-complete label of its ingredients.

A. Relational orientation as the operational task

[PROVED] Let the unknown relative orientation g in $S_0(3)$ have normalized Haar prior, let an arbitrary POVM return g_{hat} , and use full-frame chordal cost $\sin^2(\theta(g_{\text{hat}}^{-1}g)/2)$. The reference representation may contain arbitrary rotation-trivial multiplicities. For every normal state and POVM, including mixed states and invariant ancillas,

$$\begin{aligned} R_{\text{ref}} &\geq 1/(16 \langle J^2 \rangle + 8), \\ R_{\text{ref}} &\geq c_{S_03} \exp[-2 S_{\text{dir}}/3], \quad c_{S_03} = 6/(e \pi^{5/3}), \\ S_{\text{dir}} &= A_{S_03}(\rho) = D(\rho \parallel G_{S_03}(\rho)). \end{aligned} \tag{A.1}$$

The first bound follows from the spin-1 fusion matrix and a discrete Hardy inequality. The second follows from relative-entropy data processing for the covariant orbit ensemble and a global rate-distortion bound. Neither proof assumes a locally unbiased estimator, Cramer-Rao regime, purity, polarization, fixed irrep, or finite-dimensional multiplicity. This is Research Theorem W3 in THEOREMS.md and docs/global_so3_reference_risk.md.

[PROVED] The datum is the alignment of two physical frames, never orientation relative to coordinates. The loss is global over the group, not a local-sensitivity proxy. For strict integer-spin support through $j=J$, $R_{\text{ref}} \geq \sin^2[\pi/(2J+3)]$. The fermionic Finkelstein-Rubinstein sector has its separately stated projective bound.

[PROVED] Here S_{dir} is only relative entropy of rotational asymmetry. Equation (A.1) gives the necessary allocation $S_{\text{dir}} \geq (3/2) \log(c_{S_03}/R_{\text{ref}})$ where positive. It does not establish accessible classical memory, sufficiency, or a relation to S_{0b} .

[CONDITIONAL] If the reference alone undergoes stipulated isotropic rotational heat diffusion with exposure $\Gamma(T) = \int_0^T \gamma(t) dt$, the spin-1 score is attenuated exactly:

$$\begin{aligned} R_{\text{ref}}(T) &\geq (3/4) [1 - \exp(-2 \Gamma)] \\ &\quad + \exp(-2 \Gamma) / (16 \langle J^2 \rangle + 8). \end{aligned} \tag{A.2}$$

[PROVED] The regular-Gaussian-bath finite-switch theorem defines common-input exact and comparison maps and survives a finite inert ancilla. **[CONDITIONAL]** Given the open KMS GNS/Araki-Woods channel bridge, AS2 applies it to one ten-dimensional Bunch-Davies detector, pays the full operator-to-diamond factor 50, and gives normalized diamond error below 0.039 on an open nonzero-coupling box; U7 then gives $R_{\text{physical}} \geq 0.532753 \dots > 1/2$. **[OPEN]** Its exact factorized density fails microcausality under disjoint spacelike smearings and cannot be promoted unchanged to local matter; preparation, readout, stress, and backreaction are not controlled by the same action. Paper U U8a therefore remains open.

B. Capacity from localization and energy

[PROVED] For finitely many spinless nonrelativistic particles of total mass M , hard support $|x_i| \leq a$, and a ground-subtracted excitation Hamiltonian satisfying the quadratic-form inequality $H_{\text{ex}} \geq \sum_i p_i^2/(2m_i)$, every form-domain state obeys

$$\begin{aligned} \langle L^2 \rangle &\leq 2 M a^2 E_{\text{ex}}, \\ R_{\text{ref}} &\geq 1/(32 M a^2 E_{\text{ex}} + 8). \end{aligned} \tag{B.1}$$

The result includes mixed, zero-mean, and rare-tail states with rotation-trivial internal labels. It excludes intrinsic spin, relativistic or massless fields, soft localization, negative interaction energy, and nontrivially rotating light species. A separate compactness substitution is a declared proxy, not a general-relativistic body theorem. See W3a and docs/localized_orbital_reference.md.

[PROVED] More generally, sector floors ϵ_j yield $S_{\text{dir}} \leq \beta E + \log Z_H(\beta)$ only when $Z_H(\beta) = \sum_j (2j+1)^2 \exp(-\beta \epsilon_j)$ is finite. Covariance plus finite mean energy is insufficient; a bounded invariant counterexample permits increasingly accurate frame states at uniformly bounded energy. A physical theorem must derive growing sector floors from its matter model.

[PROVED] In the hard-supported massive-Skyrme hedgehog collective family on fixed de Sitter, with support radius a and minimum support lapse $N_w > 0$, the static energy and collective inertia obey

$$\begin{aligned} I[F] &\leq 4 M[F] a^2/(3N_w), \\ E_j &\geq \sqrt{3N_w} j(j+1)/2/a. \end{aligned} \tag{B.2}$$

This produces a linear large-spin floor and finite integer or projective rotational partition function. It is uniform over radial relaxation inside the adiabatic collective family (W3b).

[CONDITIONAL] The Skyrmion is therefore a concrete candidate for the capacity premise, not a complete rotating-field observer. Collective-band completeness, noncollective modes, projection error, right/isospin access, and the capacity of a marked rotating wall remain uncontrolled.

C. Certified supported profile

[PROVED] At the declared fixed-background hard-wall parameters, AU.1 is an exact-rational computer-assisted local existence theorem. It proves one solution in a certified Newton ball, strict monotonicity, a negative wall slope, and finite positive rotor inertia. The source artifact is experiments/skyrmion_newton_reduced_hessian_rounded_exact_certificate.json, SHA256 prefix c4c95db47470.

Digest strings on pages 5-7 show the first 12 hexadecimal characters. Full SHA256 values and exact artifact paths are indexed in claim_evidence_ledger.md.

[PROVED] This is local uniqueness on fixed pure de Sitter with a prescribed Dirichlet wall. It is not global uniqueness, dynamical stability, a rotating solution, arbitrary wall dynamics, or Einstein-Skyrme backreaction. **[CONDITIONAL]** It establishes that the candidate used in later calculations is more than a shooting profile; the wall remains a physical modeling input, not a harmless regulator.

D. Conserved source, master field, and electric Weyl

[PROVED] A static quadrupolar source cannot be specified by energy density alone. W3n gives the exact static even-parity bulk conservation equations and wall tractions. W3o shows that the undeformed rigid rotational Skymion stress fails this gate. This rejects the rigid source, not rotating Skymions: the centrifugal matter deformation and moving support must be included.

[PROVED] For the declared fixed-background matter-plus-moving-membrane weak form, W3z.15-W3z.16 establish a closed positive Friedrichs operator with $||A^{-1}|| \leq 100$, nonzero regular forcing, and a unique nonzero weak centrifugal deformation. The source artifacts have SHA256 prefixes 4a4e3ecd48a2 and 9edc2bc47953. Nonzero matter deformation alone does not imply nonzero exterior response.

[CONDITIONAL] Exact Hilbert-variation and shell identities, with source-bound mesh refinement, support a completed bulk-plus-moving-membrane stress at the default point. The analytic factorization is exact; closure of the selected branch is floating rather than interval certified. The safe wording is "source-bound floating conservation audit," not an unqualified certified source.

[PROVED] Once a valid conserved stress is supplied, the source, propagation, and readout maps are exact in one frozen convention. On fixed pure de Sitter, for the static $\ell=2$ normalization, regular center behavior, and no-log horizon condition,

$$\begin{aligned} A_2 \Psi &= F, & A_2 &= -d/dr[(1-r^2/R^2)d/dr] + 6/r^2, \\ F &= 8\pi G[-r^2 N \rho'/6 + r(1+4r^2/R^2)\rho/6 \\ &\quad - r p_r/2 - j + 2r \pi], & \Delta E_{rr} &= -6 \Psi(r) Y_2/r^3. \end{aligned} \tag{D.1}$$

The Friedrichs operator has a positive exact Green kernel and $A_2 \geq 6/R^2$. W3t, W3v, and W3x prove the stress-to-master identity, ideal-shell transmission, and exterior master-to-electric-Weyl reconstruction. Their artifact prefixes are 3be17612134b, c83536779d2a, and 3025d65fe825. Exact maps do not certify a nonzero completed input amplitude.

[CONDITIONAL] The completed signed estimator is stably negative, $\hat{J}$ in $[-0.0030795543, -0.0025529313]$, after including origin, correlated positive-radius, and compatible weak-wall terms. This is encouraging design evidence, not the response theorem: the rigorous residual-product interval on page 7 contains zero.

Witness boundary. If nonzero, the proposed B_W would witness non-invisibility on a fixed background. It is not the observer-capacity bound, a self-consistent geometry, or a substitute for collapse, horizon/QES displacement, or full junction data.

E. Why Paper R stopped

[PROVED] The certification architecture is exact. Outside the compact source, the fixed-background response factorizes into one amplitude $A_{\text{ext}}=J_{\text{rigid}}+B(y)$. If the symmetric weak form obeys $q(y,v)=\text{ell}(v)$ and the adjoint obeys $q(v,z)=B(v)$, arbitrary conforming trials satisfy

$$\begin{aligned} J_{\text{hat}} &= J_{\text{rigid}} + B(y_h) + R_y(z_h), \\ |A_{\text{ext}} - J_{\text{hat}}| &\leq ||R_y||_{\text{L}}(q^*) ||R_z||_{\text{L}}(q^*). \end{aligned} \quad (\text{E.1})$$

Zero exclusion requires the distance of the corrected-estimator interval from zero to exceed the product of the full primal and adjoint dual-residual bounds. This is W3z.17.

[PROVED] The bounded sprint completed every origin, bulk, and wall term. The all-strong primal and weak completed-square adjoint certificates give

$$\begin{aligned} \delta_y &< 0.785351351663998829, \quad \delta_z < 0.030892717992632714, \\ \delta_y \delta_z &< 0.024261637832088839, \\ A_{\text{ext}} &\text{ in } [-0.027341192151999246, 0.021708706437937767]. \end{aligned} \quad (\text{E.2})$$

The full interval contains zero. Under the predeclared bounded-sprint rule, the result is **INCONCLUSIVE STOP**: no nonzero exterior-Weyl theorem or equal-energy state-discrimination corollary is certified. The decision artifact prefix is bcbb4a1af96b.

[PROVED] Holding the estimator and adjoint bound fixed, zero exclusion would require $\delta_y < 0.082638613888227453$, about a 9.50x reduction from the present primal norm. The dominant primal cell is $[1/2, 11/16]$; the global coercivity floor is not the main loss.

[OPEN] Paper R is frozen as a viability memo and not promoted as a manuscript. A future attempt needs a materially different proof object, such as a better conforming primal trial, a certified Riesz solve, or a structural decomposition. More general subdivision of the present representation is deferred.

Interpretation boundary. The negative estimator is suggestive; the zero-containing theorem interval is binding. This certifies failure of the present proof representation, not physical cancellation or a no-go theorem.

F. External claim boundary

- [PROVED]** The defensible mathematical core is: global relational $S_0(3)$ risk requires asymmetry and rotational resources; named localized matter classes convert rotational capacity into energy/support cost; one hard-wall profile and its fixed-background centrifugal inverse exist; and a valid conserved quadrupolar source has exact master, transmission, and Weyl maps.
- [CONDITIONAL]** The defensible physical construction is narrower: a hard-supported Skyrmion is a plausible semiclassical realization, the regular-bath lemma supplies a conditional Bunch-Davies detector box but not a proved QFT channel or local matter, and the signed fixed-background estimator favors a nonzero exterior response. The completed rigorous interval does not certify that response.
- [OPEN]** The project has not identified S_{dir} with S_{0b} ; derived local-matter acquisition-through-readout from one action; proved a common parameter window; certified $|B_W|>0$; completed tensorial Israel matching; solved a rotating self-gravitating Einstein-Skyrme-de Sitter geometry; or shown that exterior Weyl response caps observer information. A closed-universe path integral is outside the present claims.
- [OPEN]** Novelty, priority, venue fit, and publishability are not mathematical outputs of the repository. The separate literature matrix gives the primary-source comparison. Search absence is not used as proof of priority.

G. Reproducibility map

Status	Result	Primary repository pointer
[PROVED]	Global risk/asymmetry/Casimir bounds	THEOREMS.md W3; docs/global_so3_reference_risk.md; focused tests
[PROVED]	Confined orbital capacity	W3a; docs/localized_orbital_reference.md; focused tests
[CONDITIONAL]	Named-QFT rigid-detector finite-switch channel	AS/AS2; regular-bath lemma proved, KMS GNS bridge open; factor 50 and strict error guard
[PROVED]	Supported collective sector floor	W3b; docs/supported_skyrmion_collective_spectral_floor.md
[PROVED]	Validated hard-wall profile	AU.1; exact Newton certificate listed on page 5
[PROVED]	Fixed-background source/master/Weyl maps	W3t, W3v, W3x; artifact hashes on page 6
[PROVED]	Paper R viability decision	docs/paper_r_viability_decision.md; decision hash on page 7
[CONDITIONAL]	Signed nonzero response evidence	Corrected estimator on page 6; the full interval contains zero
[OPEN]	Nonzero Weyl theorem	Needs a new primal proof object with about a 9.50-fold norm improvement

[PROVED] The complete wording guardrail, exact artifact index, and focused verification record are in docs/harlow_review_packet/claim_evidence_ledger.md. The full source comparison and its unresolved search limits are in literature_matrix.md and source_verification_log.md.

Adversarial distinctions. Absolute orientation versus relational alignment; quantum coherence versus classical record stability; S_{dir} versus S_{0b} ; measurement back-action versus gravitational backreaction; and fixed-background response versus self-consistent gravity are audited separately.

SHORT BIBLIOGRAPHY

Conceptual Neighbors

These twelve primary sources anchor the review question. The separate literature matrix audits 33 sources across observer rules, finite reference frames, clocks, de Sitter open systems, gauge-invariant response, Skyrmons, and rigorous numerics.

1. D. Harlow, M. Usatyuk, and Y. Zhao, "Quantum mechanics and observers for gravity in a closed universe," [arXiv:2501.02359](#) (2025).
2. D. Harlow, "Observers, alpha-parameters, and the Hartle-Hawking state," [arXiv:2602.03835](#) (2026).
3. V. Chandrasekaran, R. Longo, G. Penington, and E. Witten, "An Algebra of Observables for de Sitter Space," [arXiv:2206.10780](#) (2022).
4. J. De Vuyst, S. Eccles, P. A. Hoehn, and J. Kirklin, "Gravitational entropy is observer-dependent," [arXiv:2405.00114](#) (2024).
5. E. Bagan, M. Baig, and R. Munoz-Tapia, "Aligning Reference Frames Using Quantum States," [arXiv:quant-ph/0106014](#) (2001).
6. S. D. Bartlett, T. Rudolph, R. W. Spekkens, and P. S. Turner, "Degradation of a quantum reference frame," [arXiv:quant-ph/0602069](#) (2006).
7. G. Gour, I. Marvian, and R. W. Spekkens, "Measuring the quality of a quantum reference frame: the relative entropy of frameness," [arXiv:0901.0943](#) (2009).
8. B. A. Costa and Y. E. Sanchez Sanchez, "An Upper Bound for the Mass of Microscopic Clocks," [arXiv:2602.24177](#) (2026).
9. L.-Q. Chen and F. Giacomini, "Quantum Reference Fields Transformations in Linearized Quantum Gravity," [arXiv:2606.09344](#) (2026).
10. M. Fukuma, Y. Sakatani, and S. Sugishita, "Master equation for the Unruh-DeWitt detector and the universal relaxation time in de Sitter space," [arXiv:1305.0256](#) (2014).
11. H. Kodama and A. Ishibashi, "A master equation for gravitational perturbations of maximally symmetric black holes in higher dimensions," [arXiv:hep-th/0305147](#) (2003).
12. Y. Brihaye and T. Delsate, "Skyrmion and Skyrme-Black holes in de Sitter spacetime," [arXiv:hep-th/0512339](#) (2006).

Literature boundary. Existing work already covers finite observer entropy, observer-dependent gravitational algebras, degrading quantum reference frames, physical clock limits, material reference fields, de Sitter open-system dynamics, gauge-invariant response, and curved/cavity Skyrmons. The proposed distinction is their quantitative conjunction in one narrow relational $S(3)$ record model. No priority claim is made. Journal databases and terminology outside the targeted search were not exhausted.